

Load Interconnection System Impact Study Report

For Queue Project Q19-1203

Clodine 138 kV

May 17, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1203. The Interconnection Customer proposes a 372.1 MW large load (data center) facility interconnecting at the Clodine 138 kV substation in the Pennsylvania area. The request was received on May 11, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 460 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 24 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$95,544,000.

3. Network Impacts

3.1 Load Deliverability Analysis

1. Facility: Kemmerer – Trisonic 138.0 kV
Contingency: Ontelaunee Energy Center – Wyman-Gordon 138.0 kV L
Loading: 217.3% (Rating: 185 MVA)
DFAX: 60.0%
Pre-project loading: 96.5%
2. Facility: Kemmerer – Greta 138.0 kV line
Contingency: Wyman-Gordon – Ontelaunee Energy Center Substation 138.0 kV line
Loading: 188.7% (Rating: 230 MVA)
DFAX: 60.0%
Pre-project loading: 91.4%
3. Facility: Getna – Chama 138.0 kV
Contingency: Wyman-Gordon Substation – Ontelaunee Energy Ctr 138.0 kV
Loading: 187.8% (Rating: 237 MVA)
DFAX: 60.0%
Pre-project loading: 93.4%
4. Facility: S Reading Substation – Chama 138.0 kV
Contingency: Ontelaunee Energy Center – Wyman-Gordon 138.0 kV
Loading: 183.3% (Rating: 248 MVA)
DFAX: 60.0%
Pre-project loading: 93.5%
5. Facility: Wyman-Ordon – Trisonic 138.0 kV
Contingency: Chama – South Reading 138.0 kV
Loading: 174.8% (Rating: 253 MVA)
DFAX: 59.3%
Pre-project loading: 87.6%
6. Facility: Ontelaunee Energy Center – Wyman-Gordon 138.0 kV
Contingency: South Reading – Chama 138.0 kV
Loading: 173.9% (Rating: 264 MVA)

DFAX: 59.3%

Pre-project loading: 90.3%

7. Facility: Livingston Substation – Clodine 138.0 kV line

Contingency: Curecanti – Eliza Crk 345.0 kV

Loading: 167.2% (Rating: 329 MVA)

DFAX: 65.2%

Pre-project loading: 93.5%

8. Facility: Livingston – Trisonic 138.0 kV

Contingency: Curecanti – Eliza Crk 345.0 kV line

Loading: 161.8% (Rating: 353 MVA)

DFAX: 65.2%

Pre-project loading: 93.1%

9. Facility: Elk River 138/1 kV Transformer

Contingency: Eliza Creek – Ontelaunee Energy Ctr 345.0 kV

Loading: 143.7% (Rating: 516 MVA)

DFAX: 13.2%

Pre-project loading: 134.2%

10. Facility: Berthusen – Redondo 138.0 kV

Contingency: Eliza Crk – Ontelaunee Energy Ctr 345.0 kV line

Loading: 138.5% (Rating: 207 MVA)

DFAX: 7.8%

Pre-project loading: 124.4%

11. Facility: Horizon – Lake Louise 138.0 kV

Contingency: Elk Riv – Mercury 345.0 kV L

Loading: 133.4% (Rating: 242 MVA)

DFAX: 7.3%

Pre-project loading: 122.1%

12. Facility: Del Cerro – Horizon 138.0 kV

Contingency: Elk Riv – Mercury 345.0 kV

Loading: 128.9% (Rating: 231 MVA)

DFAX: 7.3%

Pre-project loading: 117.0%

13. Facility: Gray – Leake Substation 138.0 kV

Contingency: Ontelaunee Energy Ctr – Eliza Crk 345.0 kV Line

Loading: 126.1% (Rating: 182 MVA)

DFAX: 13.2%

Pre-project loading: 99.1%

14. Facility: Grizzly – Lockeford 138.0 kV line

Contingency: Elk Riv – Mercury 345.0 kV line

Loading: 121.5% (Rating: 285 MVA)

DFAX: 10.2%

Pre-project loading: 108.2%

15. Facility: Del Cerro – Grizzly 138.0 kV

Contingency: Mercury – Elk River Substation 345.0 kV Line

Loading: 119.7% (Rating: 301 MVA)

DFAX: 10.2%

Pre-project loading: 107.1%

16. Facility: Curecanti 138 kV/1 kV Autotransformer

Contingency: Curecanti Substation – Moser Generating Sta 345.0 kV line

Loading: 116.1% (Rating: 390 MVA)

DFAX: 17.7%

Pre-project loading: 99.1%

17. Facility: Eliza Crk Substation – South Reading 138.0 kV L

Contingency: Curecanti – Eliza Crk 345.0 kV

Loading: 114.3% (Rating: 310 MVA)

DFAX: 27.0%

Pre-project loading: 81.8%

18. Facility: Elk Riv 345-1 kV Autotransformer

Contingency: Eliza Crk Substation – Ontelaunee Energy Center 345.0 kV

Loading: 111.4% (Rating: 666 MVA)

DFAX: 13.2%

Pre-project loading: 104.0%

19. Facility: Dowe Flats – Curecanti 138.0 kV

Contingency: Trisonic – Livingston Substation 138.0 kV

Loading: 110.0% (Rating: 405 MVA)

DFAX: 100.0%

Pre-project loading: 18.2%

20. Facility: Fyette – Lake Louise 138.0 kV

Contingency: Mecury – Elk Riv Substation 345.0 kV

Loading: 106.4% (Rating: 297 MVA)

DFAX: 6.3%

Pre-project loading: 98.5%

21. Facility: Ontelaunee Energy Center 345/138kV TX

Contingency: Ontelaunee Energy Center – Eliza Creek 345.0 kV line

Loading: 104.2% (Rating: 456 MVA)

DFAX: 20.7%

Pre-project loading: 87.3%

22. Facility: Alliant Sbd 9106 Rockwell Cr – Dowe Flats 138.0 kV line

Contingency: Livingston – Trisonic 138.0 kV

Loading: 103.5% (Rating: 419 MVA)

DFAX: 100.0%

Pre-project loading: 14.7%

23. Facility: Berthusen – South Reading 138.0 kV

Contingency: Ontelaunee Energy Ctr Substation – Eliza Crk 345.0 kV

Loading: 103.3% (Rating: 242 MVA)

DFAX: 7.8%

Pre-project loading: 91.2%

24. Facility: Ash Street Station 223 – Cromby 345.0 kV

Contingency: Tombstone – Marcus Hook Energy Lp 500.0 kV

Loading: 101.5% (Rating: 1,036 MVA)

DFAX: 5.8%

Pre-project loading: 99.4%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Kemmerer – Trisonic 138.0 kV	Rebuild line with higher-capacity conductor	\$1,375,000	100% (\$1,375,000)
Kemmerer – Greta 138.0 kV line	Rebuild line with higher-capacity conductor	\$2,094,000	100% (\$2,094,000)
Getna – Chama 138.0 kV	Rebuild line with higher-capacity conductor	\$2,015,000	100% (\$2,015,000)
South Reading Substation – Chama 138.0 kV	Rebuild line with higher-capacity conductor	\$2,992,000	100% (\$2,992,000)
Wyman-Ordon – Trisonic 138.0 kV	Rebuild line with higher-capacity conductor	\$1,421,000	100% (\$1,421,000)
Ontelaunee Energy Center – Wyman-Gordon 138.0 kV	Rebuild line with higher-capacity conductor	\$2,515,000	100% (\$2,515,000)
Clodine Substation – Livingston 138.0 kV	Rebuild line with higher-capacity conductor	\$4,965,000	100% (\$4,965,000)
Livingston – Trisonic 138.0 kV	Rebuild line with higher-capacity conductor	\$3,884,000	100% (\$3,884,000)
Elk River 138/1 kV Transformer	Replace transformer with higher-rated unit	\$3,879,000	100% (\$3,879,000)
Redondo – Berthusen 138.0 kV	Rebuild line with higher-capacity conductor	\$4,008,000	100% (\$4,008,000)
Horizon – Lake Louise Substation 138.0 kV line	Rebuild line with higher-capacity conductor	\$1,326,000	100% (\$1,326,000)
Del Cerro – Horizon 138.0 kV	Rebuild line with higher-capacity conductor	\$12,667,000	100% (\$12,667,000)
Gray – Leake Substation 138.0 kV	Rebuild line with higher-capacity conductor	\$8,598,000	100% (\$8,598,000)
Grizzly – Lockeford 138.0 kV line	Rebuild line with higher-capacity conductor	\$1,467,000	70% (\$1,031,000)
Del Cerro – Grizzly 138.0 kV	Rebuild line with higher-capacity conductor	\$3,879,000	70% (\$2,724,000)
Curecanti 138 kV/1 kV Autotransformer	Replace transformer with higher-rated unit	\$4,042,000	100% (\$4,042,000)
Eliza Crk Substation – South Reading 138.0 kV L	Rebuild line with higher-capacity conductor	\$608,000	100% (\$608,000)
Elk Riv 345-1 kV Autotransformer	Replace transformer with higher-rated unit	\$18,464,000	100% (\$18,464,000)
Dowe Flats – Curecanti 138.0 kV	Reconductor line	\$868,000	100% (\$868,000)
Fayette – Lake Louise 138.0 kV	Reconductor line	\$374,000	100% (\$374,000)
Ontelaunee Energy Center 345/138kV TX	Replace transformer with higher-rated unit	\$13,935,000	100% (\$13,935,000)
Alliant Sbd 9106 Rockwell Cr – Dowe Flats 138.0 kV Line	Reconductor line	\$1,230,000	100% (\$1,230,000)
Berthusen – South Reading 138.0 kV	Reconductor line	\$343,000	100% (\$343,000)
Ash Street Station 223 – Cromby 345.0 kV	Reconductor line	\$6,110,000	3% (\$186,000)
	Total allocated to project		\$95,544,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.